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Full Length Article

From preparedness to action: Synthesising insights on robot usage in post-Earthquake search operations

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ABSTRACT

Human life is invaluable, and timely efforts are crucial to rescue individuals trapped under debris following an earthquake. To identify opportunities for improving post-earthquake search and rescue (SAR) robotics, we get insights through four different sources: (i) A literature review of disaster robotics and victim psychology, (ii) A public survey on earthquake awareness and their view of rescue robots, (iii) Semi-structured interviews with first responders, and (iv) Responses from GenAI chatbots which were prompted to assume the role of expert rescuers. The triangulated analysis show that there are challenges in mobility, autonomy, communication, situational awareness, and human-robot collaboration. The public respondents showed high acceptance of robot-assisted rescue and prioritised survivor detection, sensing, and communication as essential functionalities of robots. The experts expressed limitations in current equipment, the need for improved victim localisation, and interest in XR-based training and robot-assisted debris handling. The GenAI chatbots highlighted structural risk assessment, multi-sensor fusion, and supervised autonomy. Therefore, this study identifies critical robot features, outlines multi-modal interaction requirements, and highlights gaps in current SAR practice. These findings offer robot design directions for developing effective, trustworthy SAR robots, which can be integrated to future disaster-response workflows.

1. Introduction

The advancements in technology have led to developments in the field of robotics, such that they are used for search and rescue operations post-disasters. To analyse the past, current status, and foresee the future, data has been collected from four different sources for analysis in this work. We begin our discussion from the need and usage of robots for post-earthquake search and rescue operations.

1.1. The need for robots

The search and rescue missions conducted after the disaster are complex and hazardous [1]. The natural events include earthquakes, hurricanes, and floods, and man-made activities include terrorism and accidents. Using robots will improve the first respondents' search and rescue efficiency and safety. The robot can access zones that are dangerous or otherwise inaccessible for the first respondents to access. The robots fitted with sensors can be used for survivor detection, mapping, and delivering aid to victims. Integrating robots into the search and rescue process enables an efficient response to disasters.

Several factors contribute to the use of robots for post-earthquake search operations [2]. They provide safety to the first respondents in unstable structures, debris, or hazardous environments. They cover larger areas under the rubble quickly and systematically by using sensors and cameras. They collect critical information from places difficult for first respondents due to structural damage. They help make decisions by collecting real-time data. By using robots, the search process can be made faster and more reliable.

The work in [3] demonstrates the need for robots to inspect structures after disasters. Using robots enables efficient data collection and travel through hazardous paths. The experts can analyse data and make decisions by examining the structure of the collapsed building. A safer and more efficient assessment can be made using robots for structural inspections.

In the works of [4], the opinion of using the *Spot* quadruped robot for post-earthquake search operations received an overall positive response among the rescuers. They expected the robot to work autonomously to search and gather site information, as it would be more efficient. Consequently, the rescuers expressed a preference for robotic assistance operating in autonomous mode.

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1.2. The usage of robots

In the recent past, we find that robots are being used for SAR. The robots capable of being used in search and rescue operations before 2007 are detailed in [5]. These include snake robots such as ACM R3 [6] and GENBU [7], Unmanned Mobile Rescue System (UMRS) series search and rescue robots, and nuclear radiation operation robots in Japan such as RESQ-A and RESQ-B. Apart form this, it presented SAR robots made in the USA and Japan, that were used on field of disasters. Future search and rescue robots will rely on integrating advanced technologies such as AI, multi-sensor fusion, and networked multi-agent systems. Insights into the future of search and rescue robots can be found in [8]. It suggests the need for greater autonomy, improved communication, and enhanced sensory capabilities to operate effectively in challenging environments. The SAR robots can be evaluated using performance tests and an index system establishment [9,10]. The existing HRI medium hinders collaboration between tele-operators and robots [11]. The robots and the situations they were used in after disasters are briefly discussed:

- *Great Hanshin Earthquake (1995):* Post the earthquake in Kobe, Japan, the challenges faced by the first respondents emphasized the need for search and rescue robots. After this event, the RoboCup Rescue Simulation project was started that developed robots for search and rescue operations [12,13].
- *September 11, 2001 Attacks:* Robots of several sizes were deployed for the search operations after the World Trade Center collapse. These robots reached the hazardous regions inside the debris and helped find survivors beyond human visibility [14].
- *Tōhoku Earthquake and Tsunami (2011):* The robots played a critical role in the search operations after this event. They were used to assess unsafe areas and inspect buildings that were damaged [15].
- *Mexico City Earthquake (2017):* Post the earthquake, the Mexican Red Cross sought the help of Carnegie Mellon University to use their snake robot to assist the search operations. It navigated through debris that were inaccessible for the rescuers [16].
- *Western and Central Europe floods (2021):* Post the flood, aerial robots were used for mapping the environment. An operator used a head-mounted display for the first-person view, where the camera was attached to the drone [17].

These events highlight the critical role of robots in enhancing the efficiency and safety of post-earthquake search and rescue operations. In the Niigata Chuetsu Earthquake, Japan (2004), snake robots were used for the first time. The commonly used robots include unmanned ground vehicles (UGVs), biomimetic robots, aerial robots, marine robots, and hybrid robots. A detailed discussion can be referred to in [18].

1.3. Goals of the study

The primary objective of this study is to understand the current perceptions and expectations related to earthquake awareness and the use of robots in post-earthquake search and rescue operations. Specifically, the study aims to:

- Understand the trends through current literature and victim perspective from the published works.
- Evaluate emotional responses associated with earthquakes and robot usage from the public.
- Identify the pain points of the first respondents and understand their needs and functions required in SAR robots.
- Understand the information from GenAI chatbots to enhance robotic interventions.
- Perform data triangulation and identify common pain points, features of the robot, and multi-modal interaction.

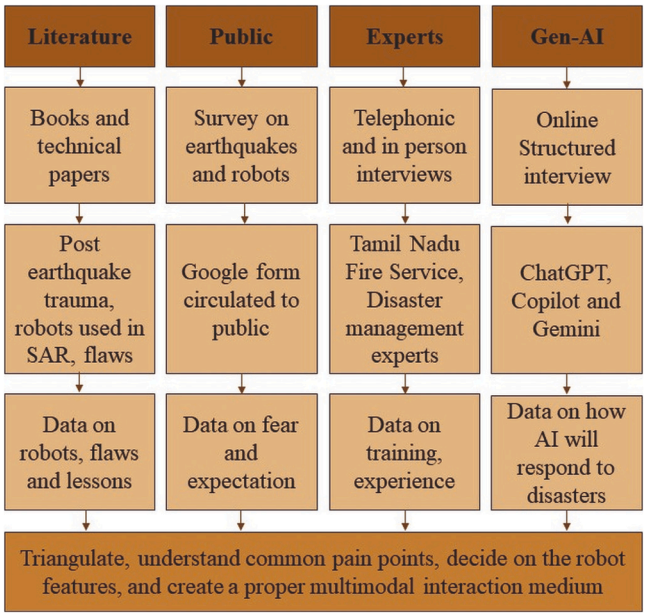


Fig. 1. The overview of the data collection process followed in this work. There are four sources namely, the literature, public opinion, expert opinion and Gen-AI opinion.

1.4. Overview of the data collection process

The data collection process was conducted through a systematic review of the existing literature, getting public and expert opinions, and getting GenAI chatbots perspective. This procedure ensures that the data collection process incorporates both quantitative and qualitative elements. This helps understand the challenges and opportunities in the post-earthquake search operation by using robots to assist the first respondents. An overview of the process is presented in Fig. 1. This study aims to understand the common pain points, decide the features for the robot, and create multi-modal interaction for the robot.

1.5. Data analysis method

The literature survey was done using the information from [18–20]. The survey responses were analysed using descriptive statistical techniques. The frequencies, percentages, and Likert-scale distributions were used to characterise earthquake awareness, emotional reactions, and acceptance of robot-assisted rescue. Qualitative data from expert interviews and GenAI chatbots responses were analysed to identify the gaps and challenges in the usage of SAR robots. The data from the four sources were triangulated and presented in this work.

2. Literature review

In this section, we perform a literature survey using three sources ([18–20]) on disaster robotics and understand the victim perspective of the earthquake. This includes the survival time and the psychology of the victims.

2.1. Related works on robot usage for search and rescue

2.1.1. AI And robotics in disaster management

The work [19] provides details about the application of artificial intelligence and robotic technologies in pre-disaster preparedness and post-disaster management. It highlights seismic events, early warning systems, historical data acquisition, predictive models, and the role of ground-based robotic platforms for survivor detection within debris.

Unmanned aerial vehicles are employed for high-resolution aerial mapping and damage assessment.

2.1.2. Disaster robotics

The motivation for the work *"Disaster robotics"* [18] emerges from significant disasters such as the Oklahoma City bombing and the Kobe earthquake. Robots can substantially improve situational awareness and operational efficiency, and their effectiveness depends on the particular disaster context.

Case Studies and Lessons Learned. Kobe Earthquake (1995): Robots were used in the search and rescue operation. This case emphasised the importance of real-time data processing and the need for robots to work with human teams.

World Trade Center (2001): Robots were employed to search for survivors in the debris of the World Trade Center. Deployment exposed the lack of communication and navigation because many robots could not operate appropriately in the debris.

Tōhoku Earthquake and Tsunami (2011): Robots (PackBot and Quince) were used for data extraction which were reported to emergency responders. The case showed the role of pre-disaster planning and training.

Fukushima Daiichi Nuclear Accident (2011): Robots were used to measure radioactivity and inspect damaged structures. The case highlighted the need for more autonomy and navigation in high-radiation environments.

2.1.3. Disaster robotics: Results from the impact tough robotics challenge

The work [20] began after the Great Hanshin-Awaji Disaster in 1995. The ImPACT Tough Robotics Challenge (ImPACT-TRC) focused on developing durable and bendable robotic platforms to enhance safety and security in extreme conditions. The program provided mechanisms for dealing with search and rescue operations, disaster preparedness, and component technology advancements.

2.2. Perspective of the victims

The victims are the primary targets during search and rescue operations. The survival time and psychology of the victims need to be studied.

2.2.1. Survival time

The works of [21] highlight that survivors inside collapsed buildings can live for 5–6 days and in some cases up to two weeks. Several instances are highlighted where survivors were rescued after extended periods, showing that humans can survive trapped inside collapsed buildings when they have access to food and/or water.

2.2.2. Psychology of the victims

The survivors often face psychological issues such as acute stress reactions, anxiety, depression, and post-traumatic stress disorder (PTSD). Their daily life post-earthquake is affected by concerns of safety and instability. Additionally adverse effects, post-traumatic growth (PTG) is observed in some survivors [22].

3. Public perception

It is important to understand the public opinion on how they perceive earthquake and the usage of robots for post earthquake search and rescue operations. A questionnaire was prepared and was circulated in a Google form. The form had a title *"Survey to Understand General Public Opinion on Earthquake Awareness and Rescue Robots"*. It had a declaration statement and 15 questions that the participant had to fill. About 238 participants filled up the form. The data collected was processed and presented as follows.

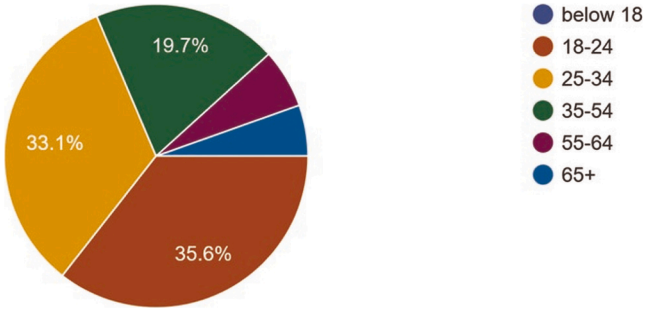


Fig. 2. The pie chart illustrating the distribution of respondents across different age groups, showing the overall demographic spread of participants in the survey.

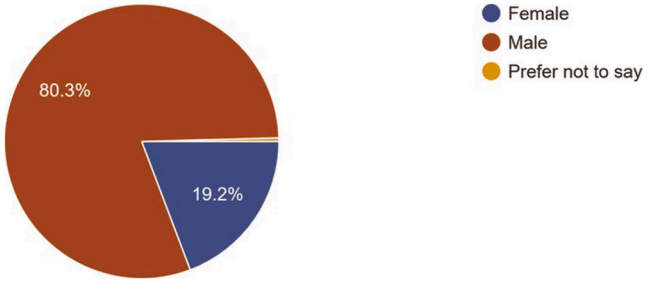


Fig. 3. The pie chart illustrating the gender composition of the survey sample, highlighting the representation of different gender identities among the respondents.

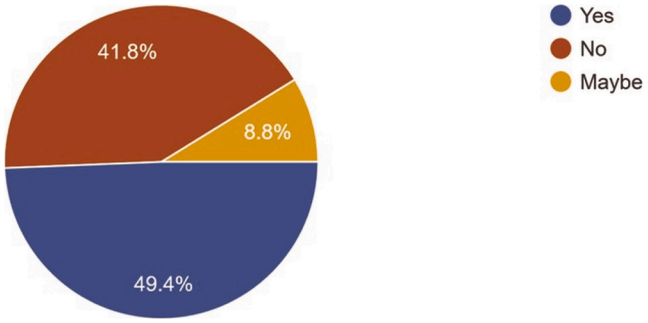


Fig. 4. The pie chart shows whether respondents have previously encountered an earthquake, providing an overview of their past exposure to seismic events.

3.1. Participant demographics and earthquake experience

The survey received responses predominantly from younger adults, with the largest age groups being 18–24 years (35.7%) and 25–34 years (33.2%), together accounting for nearly 69% of participants (see Fig. 2). Other age groups included 35–54 years (19.3%), with fewer respondents aged 55 and above. Regarding gender distribution, males constituted the overwhelming majority at 80.3%, with females representing 19.3%, and a small fraction opting not to disclose their gender (Fig. 3).

When asked about previous exposure to earthquakes, responses were fairly balanced: approximately 49.2% of participants reported having experienced an earthquake, 42% had not, and 8.8% were unsure or ambiguous about their experience (Fig. 4).

3.2. The results of the survey

3.2.1. Earthquake awareness and reaction

1. *Perception to the earthquake:* How afraid are you of earthquakes? The answer to this question helps us understand how the public is prepared and feels about the earthquake.

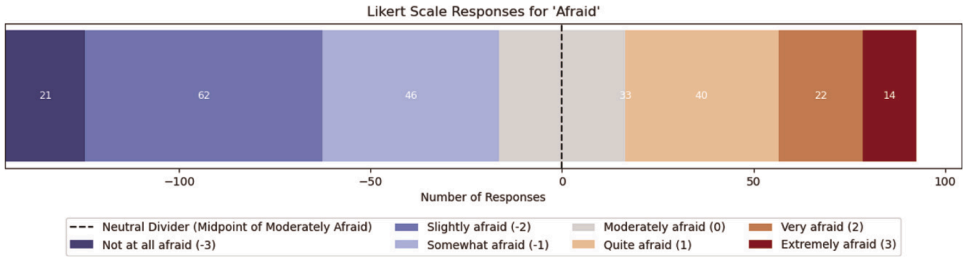


Fig. 5. The Likert scale shows the public perception to the earthquake. The question asked was: *How afraid are you of earthquakes?*.

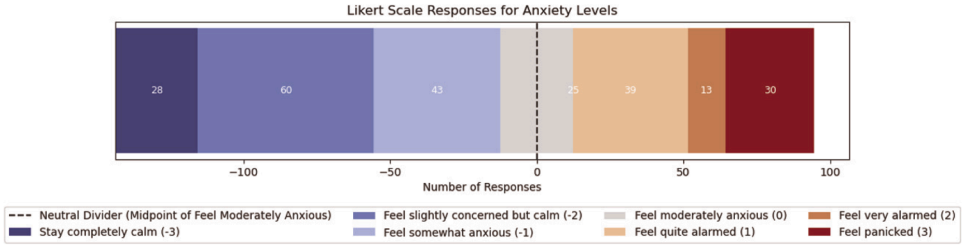


Fig. 6. The Likert scale shows the public reaction to earthquake. The question asked was: *What would be your reaction when you feel an earthquake?*.

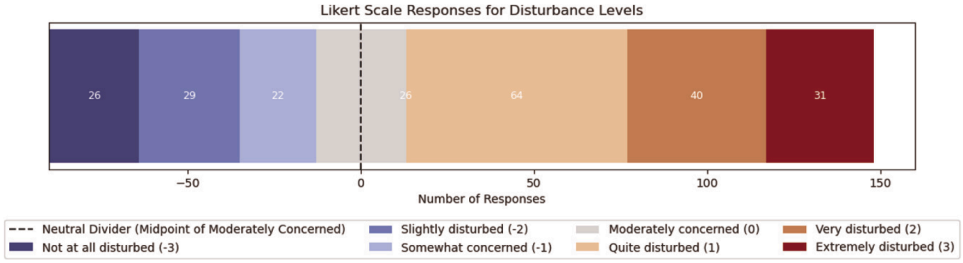


Fig. 7. The Likert scale shows the public reaction to the damaged building. The question asked was: *What is your reaction to the post-earthquake damaged buildings?*.

The survey quantified the participant's fear on a Likert scale to determine their emotional state. Results shown in Fig. 5 demonstrate that most respondents have been calmer. In addition, future studies could reveal the conditions that give rise to these emotional imbalances and will help to more carefully assess fear in similar contexts.

2. *Reaction to earthquake:* What would be your reaction when you feel an earthquake? The answer to this question helps us understand how the public reacts to the earthquake.
   The survey aimed to evaluate anxiety levels using a Likert scale. Results in Fig. 6 show a consensus of many participants who did not show extreme emotions. These results indicate that although most participants reported high to moderate intensity of anxiety, a substantial proportion reported high intensity of anxiety.
3. *Reaction to the post-earthquake damaged buildings:* What is your reaction to the post-earthquake damaged buildings? The answer to this question helps us understand the trauma of the public after the earthquake.
   The survey aimed to evaluate disturbance levels using a Likert scale. Results in Fig. 7 indicate a substantial proportion of the subjects reported severe levels of disturbance. Further investigation is necessary to clarify the forces shaping such perceptions and how they can be overcome, particularly for those with high levels of disturbance.
4. *Reaction when they are stuck in the damaged building:* What would be your thoughts when you are stuck inside a collapsed building? The answer to this question helps us understand the public's post-earthquake thought process.
   The plot in Fig. 8 illustrates the impact of a 7-point Likert scale fear magnitude questionnaire. These results show a clear directionality toward fearfulness in the sample.

3.2.2. Robot awareness

1. *Reaction to the robot that comes to the rescue:* How afraid are you when a robot rescues you inside a collapsed building after an earthquake? The answer to this question helps us understand what the public feels about using rescue robots.
   The questionnaire evaluated the fear scores on a 7-point Likert scale. These results point to a general feeling of calmness in the sample.
2. *Opinion on using robots to rescue:* What is your opinion on using robots for search operations inside collapsed buildings? The answer to this question helps us understand how the public accepts robots as a part of the rescue process.
   The survey aimed to quantify the acceptance of an intervention as a support for a 7-point Likert scale. The findings shown in Fig. 10 suggest a high acceptance of the proposal. The results point to the majority of the public accepting the usage of robots for the search and rescue process.
3. *Type of robot preferred:* What type of robots do you prefer encountering when you are inside a collapsed building after an earthquake? The answer to this question helps us understand the form of robot that the public expects.
   The question seeks the preference between bio-inspired and machine-like designs on a Likert scale. Results shown in Fig. 11 provide a mixed corpus of responses with a slight majority of bio-inspired design. The results show a consistent tendency towards bio-inspired designs, but the high proportion of neutral responses suggests a low preference level for many respondents. Additional research is required to gain a deeper understanding of these satisfying preferences and the factors that motivate the participants' choices.

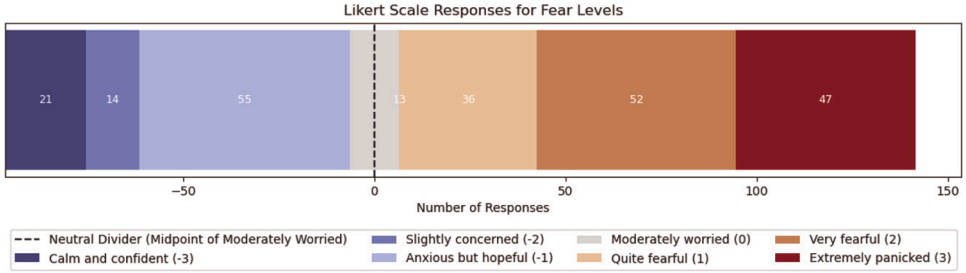


Fig. 8. The Likert scale shows the fear levels of the public when they are stuck in the damaged building. The question asked was: *What would be your thoughts when you are stuck inside a collapsed building?*.

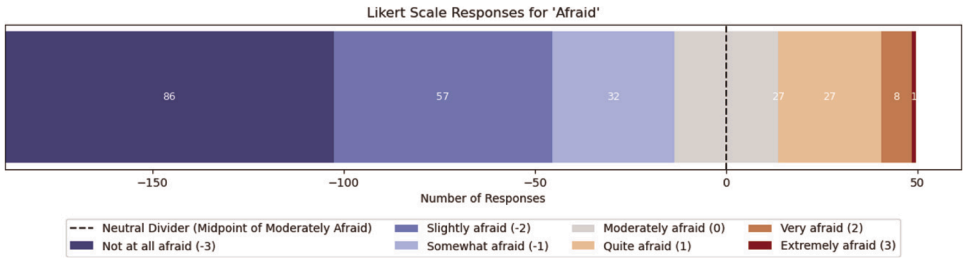


Fig. 9. The Likert scale shows how the public reacts when a robot comes to rescue them. The question asked was: *How afraid are you when a robot rescues you inside a collapsed building after an earthquake?*.

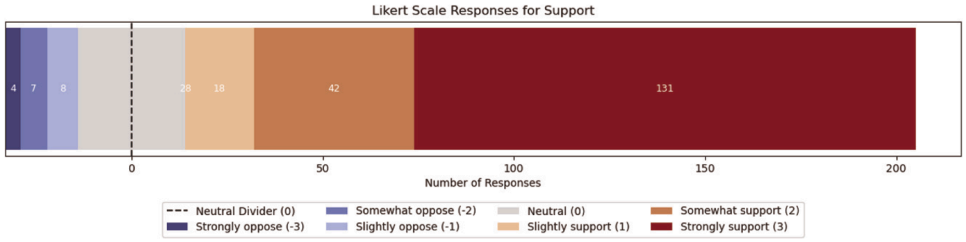


Fig. 10. The Likert scale shows the public opinion of using robots for search operations. The question asked was: *What is your opinion on using robots for search operations inside collapsed buildings?*.

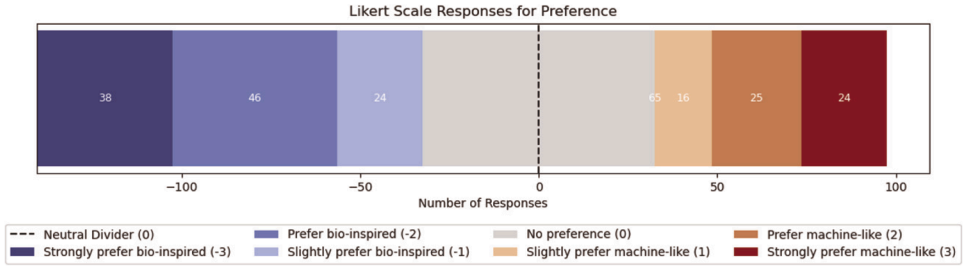


Fig. 11. The Likert scale shows the robot preferred by the public. The question asked was: *What type of robots do you prefer encountering when you are inside a collapsed building after an earthquake?*.

4. *Choice between humans and robots:* Whom do you rely on to save you when you are stuck inside a collapsed building? The answer to this question helps us understand whom the public relies on when stuck in a collapsed building.
   The survey assessed preference between human and robot use in a Likert scale format. Results shown in Fig. 12 demonstrate a broad and heterogeneous response effect, strongly dominated by an affective preference for human assistance. Results demonstrate a robust human reliance bias but many neutral responses, suggesting that many subjects are undecided. However, more studies are needed to explore the factors contributing to reliance preferences and overcome heterogeneous prejudices.

3.2.3. Function of robots

1. *Speeding up the search operation:* How effective do you feel the rescue robots will speed up the post-earthquake search operation? The answer to this question helps us understand the public opinion on whether using robots will be effective.
   The inputs from the study was plotted in a Likert scale. Results shown in Fig. 13 indicate a substantial number of respondents rated the usage as very effective. These results demonstrate a spectrum of effectiveness perceptions, with a clear expectation of positive responses. Variation in response could also be due to differences in experience, expectations, or context. Further work is required to determine the predictors of these beliefs and agree on how best to improve

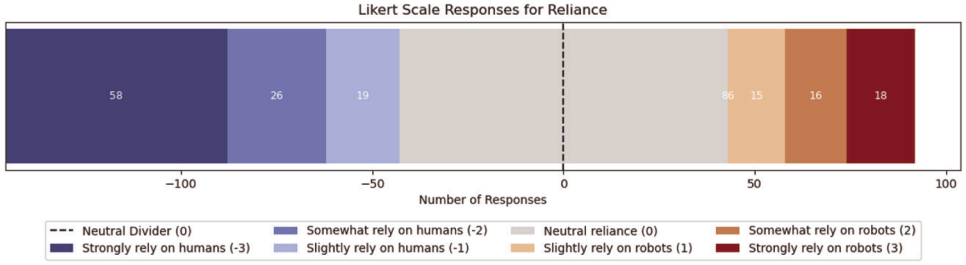


Fig. 12. The Likert scale shows whom the public rely on to save them when they are stuck inside a collapsed building. The question asked was: *Whom do you rely on to save you when you are stuck inside a collapsed building?*.

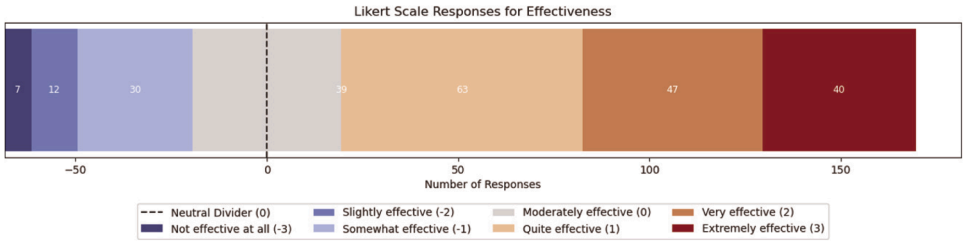


Fig. 13. The Likert scale shows the effectiveness of the usage of rescue robots. The question asked was: *How effective do you feel the rescue robots will speed up the post-earthquake search operation?*.

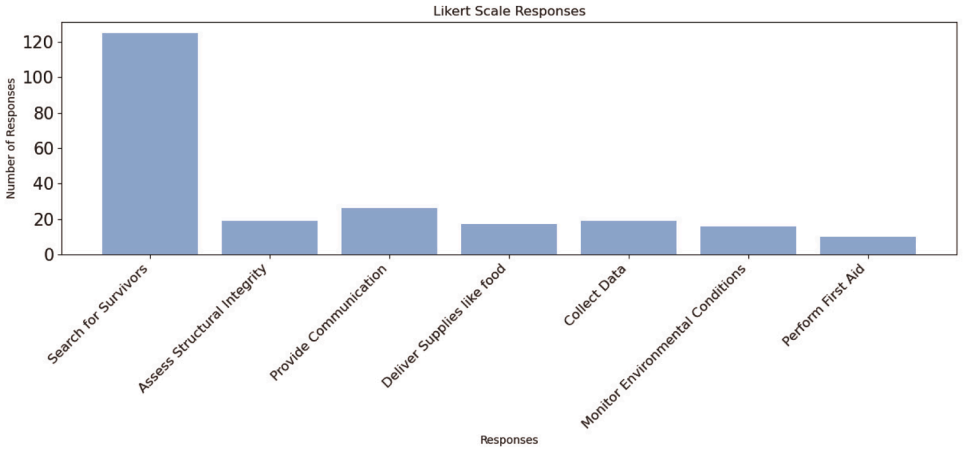


Fig. 14. The essential operational functions required to be executed by the robotic systems. The question asked was: *What are the most critical tasks a rescue robot should perform post the earthquake?*.

effectiveness, particularly for those parts where participants rated lower levels of impact.

2. *The most critical task:* What are the most critical tasks a rescue robot should perform post the earthquake? The answer to this question helps us understand the priority of tasks the public expects the robot to do. This will help to design the robot by giving importance to these features.
   The response is shown in Fig. 14. This chart places much emphasis on *"Search for Survivors"* while giving much lesser importance to other tasks.
3. *Inspiration from nature:* Is inspiration from nature (like animals) essential to build a rescue robot? If people accept bio-inspiration, it will save time when designing robots.
   The results from the survey varied, showing (in Fig. 15) perceptions of the importance of the evaluated factor, with a majority of positive ratings. This distribution thus shows that most respondents see both factors as necessary, but a significant minority feel neutral or even less critical about the factor. These results indicate that further research should be conducted to understand why some consider this factor essential and not others.
4. *Speeding up the search operation:* According to you, what is an essential feature of an ideal rescue robot from a survivor's point of view? This will help give preference to the features from the survivor's point of view.
   The bar chart in Fig. 16 displays the responses for each factor in a given context. This distribution clearly shows a preference for operational functionality features like sensors and communications over secondary factors, such as robustness and autonomy.

4. Expert insights

The opinion of the experts would give a new dimension to the usage of robots. A questionnaire was prepared, and semi-structured interviews

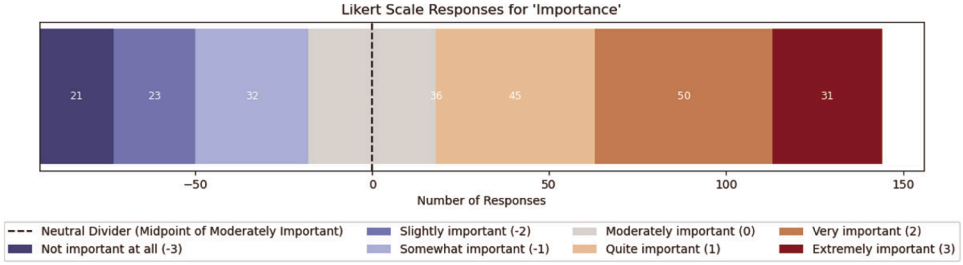


Fig. 15. The Likert scale shows the public opinion on bio-inspiration for building robots. The question asked was: *Is inspiration from nature (like animals) essential to build a rescue robot?*.

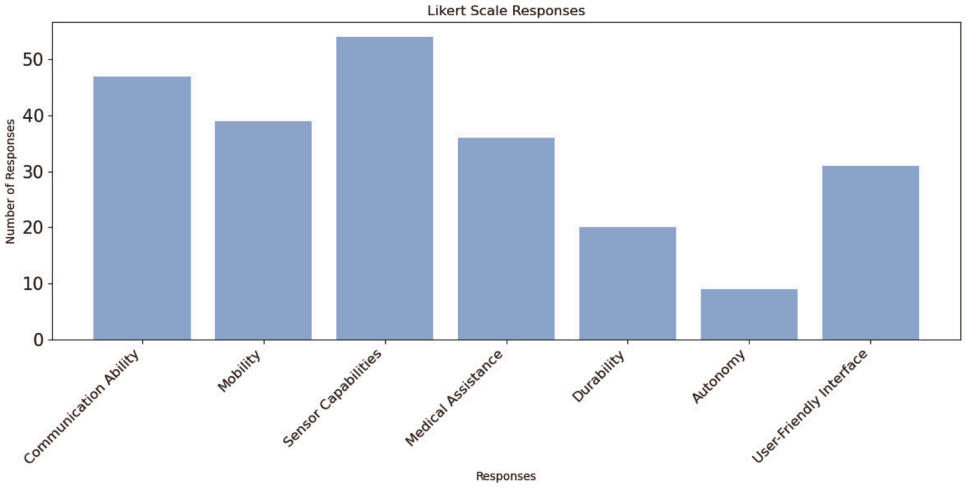


Fig. 16. The public view on essential features of an ideal rescue robot. The question asked was: *According to you, what is an essential feature of an ideal rescue robot from a survivor's point of view?*.

Table 1
Details of the experts who participated in the semi-structured interview.

Pseudo Name	Expert Information	Years in Service	Location	Current Role	Experience in search operations
F1	Fireman	3	Tamil Nadu, India	Fire extinguishing and human rescue operations during disasters.	Yes
F2	Fireman	1.5	Tamil Nadu, India	Fire extinguishing and human rescue operations during disasters.	Yes
F3	Fireman	25	Assam, India	Defence fire service	No
D1	Disaster Management Expert	13	Delhi, India	Develop policies and robots for post-earthquake search operations.	Yes
F4	Fireman	6	Tamil Nadu, India	Fire extinguishing and human rescue operations during disasters.	Yes

were conducted with the first four respondents and a disaster management expert. The questionnaire is presented in Section Appendix A.

4.1. Introduction

4.1.1. The details of the experts who were interviewed

The experts were interviewed based on the questionnaire prepared. The experts include firefighters and personnel from disaster management. Their details are presented in Table 1.

4.1.2. Public awareness program on earthquakes

The experts F1, F2, and F4 have conducted awareness programs for the public in Tamil Nadu, India. It includes theory classes and demonstrations of the rescue process to the people. The expert D1 has conducted earthquake awareness programs that focus on behaviour change.

4.2. Current situation

The expert F4 said that they need to assess the structural integrity of the building to prevent the respondents from falling into danger. The expert F1 reported that the electricity supply to the house must be cut off. Gas cylinders might leak, leading to fire accidents; therefore, gas leakage must be detected, and the supply must be cut off. The experts F1, F2, and F4 were involved in several collapse situations, including a building under construction, a tree falling on a weak house, and the collapse of a petrol bunk. The expert F4 has worked in situations where there was

a collapse at a stone quarry, and workers were trapped between stones. The expert F1 said that they would have the necessary items required for the rescue process, including a ladder, ropes, a jacket, a helmet, shovels, spades, fire hooks, and saws. The experts F1, F2, and F4 rush to the place of collapse in a fire engine after collecting the required materials. On the way, they wear the safety equipment and prepare an action plan. The expert F4 said that they would take control of the area with the help of local police. The expert F2 said that they would blow the siren and make an announcement. Their seniors inspect the collapsed structure and plan entry points and tools needed for the search process. The expert F4 searches in the voids of the collapsed building, asks for information from witnesses, and assesses available CCTV footage. In cities, the experts F2 and F4 use trained sniffing dogs to identify humans. The expert F2 said they would end the search process when instructed by the district fire officer. The expert D1 stated that they would continue until all the survivors or bodies were retrieved.

4.3. Current devices

The expert F1 reported using personal safety devices like jackets, helmets, and lifebuoys. They also had ladders and ropes to aid in climbing and debris removal. Expert F2 said it depends on the severity and the survivor's location. Trained sniffing dogs were used to find survivors in urban areas. The experts F1 and F4 mentioned using shovels and spades. The expert F4 used ropes to tie debris and pull it using the fire engine motor. The expert D1 reported using ground-penetrating radars, probes with cameras, and drones. The experts F1 and F4 needed a system or device to help tie ropes and lift debris. Expert D1 said the search and rescue process should be automated and that separate robots should be used for search and rescue. The expert F3 indicated a willingness to see recent technological advances to help their team. The expert F1 said some proactive people come forward and disturb the first responders. The experts F1, D1, and F4 reported that the pinpoint location of survivors and their condition is required. The expert F4 reported that moving heavy debris is a significant challenge.

4.4. Use of robots

When the usage of robots were questioned, none of the experts has worked alongside robots so far, but all of them would like to have robots assist in the rescue process. The expert F1 prefers a humanoid robot so that they can find a companion during the process. The expert F2 wants a robot or device that can find people by touch, useful in dark and smoky environments. The expert F3 intends to use robots, but has not encountered any that can help. The expert D1 wants robots to assist in search operations and sense victims to give feedback to the team. The expert F4 wants a robot that can detect victims and their status and interact with trapped victims.

4.5. Concluding remarks

The expert F1 wanted the robot to be stable, fireproof, waterproof, capable of lifting debris beyond human capability, tying knots, and recovering small, valuable objects. The expert F2 is interested in getting trained through virtual reality headsets. The expert F3 is willing to adapt to new technologies. Expert D1 feels there is a need for better human-robot communication. The expert F4 needs a portable debris breaker, bright portable lights, and motors to pull debris. All the experts are willing to collaborate with technologists to build efficient rescue robots.

5. Genai chatbots perspective

The recent advancements in artificial intelligence have resulted in the deployment of GenAI chatbots-driven search tools in disaster response scenarios. Incorporating the perspective of GenAI chatbots systems is crucial to evaluating their impact on improving the efficiency and effectiveness of search operations in post-earthquake environments. In this work, the GenAI Chatbots interviewed were ChatGPT, Gemini, and Copilot. They were asked to assume the role of a first respondent during post-earthquake search and rescue operations. The questions asked were identical to those posed to the experts. We initiated each chatbot session with the following standardized prompt:

I am going to interview with you. I assume you are an expert first responder rescuer for post-earthquake search operations. Based on your experience, answer the questions that I ask.

The questions asked to the GenAI chatbots are similar to the questions in Section Appendix A.

5.1. Current situation

5.1.1. Current situation by chatgpt

The current situation includes safely clearing debris using equipment such as hydraulic spreaders and cutters. It emphasises that heavy machinery should be used for large-scale debris removal and that robotic systems and drones should be used in unsafe areas. Successful communication among members of the Incident Command System (ICS) is essential. Secondary hazard management such as gas leaks, fire hazards, and post-seismic shaking is also required.

5.1.2. Current situation by copilot

The first stage consists of gathering information about the site, the degree of damage, the number of potential victims, and possible emergencies. Emergency responders such as paramedics, police, and structural engineers are contacted. Resource mobilisation is emphasised, including thermal imaging cameras, acoustic sensors, search dogs, and medical equipment. Establishing a command post provides control over communication and logistics. Rapid risk assessment is used to evaluate existing dangers.

5.1.3. Current situation by gemini

Gemini recollects experience in urban search and rescue efforts in collapsed buildings. It outlines collecting and securing unsteady collapsed structures and locating victims with listening equipment, search dogs, and cameras. It highlights the need to extract trapped victims and to evaluate and treat injuries immediately. It emphasises specialised training, teamwork, and coordination among rescuers.

5.2. Current devices

5.2.1. Current devices by chatgpt

This discusses tools used in post-collapse search and rescue. Critical equipment includes hydraulic tools, airbags, drones for aerial reconnaissance, ground-penetrating radar, and acoustic sensors. Robotic devices allow remote exploration of dangerous locations and deliver real-time data to support situational understanding.

5.2.2. Current devices by copilot

The devices include search dogs, thermal cameras, acoustic sensors, bendable search cameras, and ground-penetrating radar (GPR). Life detection apparatuses detect cardiac rhythm and subtle signs of survivors.

5.2.3. Current devices by gemini

Gemini highlights the interaction between fire services and technology engineers. Field testing is needed to evaluate designs. Data exchange and analysis help identify opportunities for improvement. Collaborative research and development projects can enhance earthquake rescue technology.

5.3. Use of robots

5.3.1. Use of robots by chatgpt

There is an increasing need for robot applications in SAR operations. Ground robots can navigate through rubble and tight spaces and can

Table 2
Data Triangulation of Literature Review, Public Perception, Expert Insights, and GenAI Chatbots Perspective.

Aspect / Theme	Literature Review	Public Perception	Expert Insights	GenAI Chatbots Perspective
Understanding of Robot Usage in Disaster	Reviews historical and current use of robots	Awareness is moderate; acceptance conditional on safety	Emphasises practical benefits and operational challenges	Highlights AI's role in predictive analytics and data integration
Technological Limitations and Challenges	Reports mobility, autonomy, communication issues	Scepticism due to perceived unreliability	Points out on-field constraints	Notes challenges in transparency and trust
Expectations for Robot Features	Recommends enhanced mobility and sensors	Public expects resilient, easy-to-use robots	Experts request robust communication and multi-modal interaction	Suggests AI augmentation of robot capabilities
Role of Human-Robot Interaction	Integration of interaction is an area for improvement	Emphasises trust and effective use of robot feedback	Highlights seamless communication and interpretable outputs	Requires transparency for informed decisions
Use of AI and Data-driven Approaches	Notes emerging use of AI	Public supports technological advancements	Experts acknowledge AI's potential with oversight	AI supports data processing and prediction
Overall Sentiment Towards Disaster Robotics	Optimistic but acknowledges gaps	Positive but cautious	Supportive with practical concerns	Positive but warns about trust-related challenges

be fitted with cameras and microphones. Fixed-wing and multi-rotor systems provide aerial remote sensing. Partial autonomy is necessary with user supervision, and communication challenges remain.

5.3.2. Use of robots by copilot

Robots with sensors, cameras, and thermal imaging can identify victims and hazards. They can perform site surveys and navigate confined and unstable environments. Robots monitor structural conditions and relay required data. Integration with human teams enhances decision-making and safety.

5.3.3. Use of robots by gemini

The application of robots is valuable for improved performance and speed of rescue operations. Robots equipped with sensors and cameras enhance situational awareness and victim location. They can perform tasks such as structural integrity checks and supply delivery. Integration between engineers, scientists, and rescue team members must inform robot design.

6. Discussion

The literature review, public opinion surveys, the viewpoints of the first responder community, and the GenAI chatbots have all been discussed. Analysing these sources, we can identify some commonalities, pain points, and areas for improvement in the effectiveness of robotic systems in disaster situations.

6.1. Data triangulation

The data triangulation was used as a methodological tool to find similarities. This approach combines data collected from literature reviews, public surveys, expert interviews, and interactions with GenAI chatbots. This is tabulated in Table 2.

6.2. Common pain points

Although the information is derived from four sources, there are recurring difficulties that robots will encounter in disaster management:

1. *Technological Limitations:* Challenges related to mobility, autonomy, and communication capabilities hinder robots from operating effectively in hazardous environments.
2. *Human-Robot Collaboration:* Despite public acceptance of robots, the importance of human control remains. Experts emphasised the necessity of training and practice.
3. *Integration into Existing Frameworks:* Integrating robotic devices with current emergency response systems is an issue. Responders must learn how to use new technologies.
4. *Trust and Acceptance:* Public acceptance indicates the need to build trust in robotic technologies.
5. *Psychological Factors:* Emotional responses in disaster-affected populations highlight the need to develop robots that can address victims' feelings.

6.3. Multi-Modal interaction for the robot

Multi-modal interaction platforms can effectively contribute to human-robot interaction in disaster response. These are some of the critical interaction modalities that could be beneficial.

1. *Voice Communication:* From a control perspective, a robot's competence to identify what and how to resist a speech command can be understood to support intuitive communication. Voice recognition systems may be employed for real-time use in inter-human robot communication, an essential tool for sending command or information reception.
2. *Visual Displays:* Robots equipped with screens or projectors can display visual contents (maps, status, video streams received by the robot sensors). This visual information may assist human operators in arriving at informed decisions as quickly as possible.
3. *Touch Interfaces:* The user can control the robots directly through touch screens or haptic interfaces. It is very effective for training or when fine control is necessary in adverse circumstances.
4. *Gestural Control:* The control of a robot/machine by hand gestures or body movement can be realized using gesture recognition technology. This could be beneficial when voice communication is not feasible, e.g., in a noisy environment or for personnel wearing protective gear.
5. *Augmented Reality (AR):* AR's functionality includes generating digital data on the outside world to increase responders' situational awareness. AR glasses, for instance, may present robot-generated data, such as a robot's performance and possibly structural integrity analysis, as well as where people got trapped.
6. *Mobile Applications:* Developing mobile applications for robots used by first responders will enable remote supervision and control of robotic systems. These apps could provide real-time information, alerts, and control features, helping to manage and coordinate multiple robots efficiently during a rescue operation.
7. *Social Cues and Emotional Feedback:* Creating social behavior (e.g., expressions or gestures) in robots enables us to promote trust and foster the emotional attachment between humans and robots. This is particularly important in high-stress situations like disasters.
8. *Data Visualization Tools:* The capability to integrate data visualization tools that render complex data in an easily understandable way for responders is essential. This should allow responders to assess their situation and determine their course of action directly from the robot's feedback.

By integrating these interfaces, the efficiency of multi-modal interactivity in human-robot interaction can be improved, and better disaster response can be achieved.

7. Contributions, limitations and future work of this study

7.1. Contributions

The work utilizes data from four sources to enable holistic understanding of the current state and challenges in earthquake-related SAR robotics. The data collected provides a solid understanding of multiple perspectives. The study highlights essential tasks that rescue robots should prioritise, including survivor search, structural integrity assessment, communication, and environmental monitoring. These would help robot designers incorporate specific features into SAR robots. The study proposes integrating tools that allow responders to assess situations efficiently through robot feedback. Several modalities are identified that would prevent situational impairment and lead to effective communication with the robot. Incorporating GenAI chatbot interviews illustrates the potential of AI to complement human responders, support decision-making, and contribute to safer, more efficient rescue workspaces. The study stresses the need for cooperation between technology developers and first responders. The collaborative development of robots enhances the robot's functionalities. By assessing public views on bio-inspiration and robot task priorities, the research acknowledges societal factors in robot adoption, ensuring future technologies align with user expectations.

7.2. Limitations

This study offers an examination of the role of robotics in post-earthquake search and rescue operations. The findings are based on literature, surveys, expert interviews, and GenAI chatbots, which may not fully capture real-world disaster environments. The technical capabilities and constraints of current robotic platforms remain significant challenges. Public perception data collected via online surveys introduces sampling bias. GenAI chatbots responses are restricted to their training data. The integration of robotic systems within established emergency workflows needs in-field validation and iterative refinement.

7.3. Future works

Future research will focus on developing and implementing the key technological gaps highlighted in this study, particularly improving robot mobility, autonomy, and communication systems in complex disaster environments. Collaboration between engineers, first responders, and policymakers should help integrate robotic systems into emergency response frameworks. Dedicated training programs are needed to ensure human-robot cooperation. Public engagement should be increased to utilise robotic assistance during disaster response. Multi-modal human-robot interfaces can enhance situational awareness. AI-assisted technologies should be evaluated in real-world scenarios to advance robotics for earthquake search and rescue missions.

8. Conclusion

This study integrates information from literature, public perception, expert interviews, and GenAI chatbot perspectives to provide an understanding of robotic assistance in post-earthquake search and rescue. From these four sources, it was identified that robots should be capable of navigating confined rubble, detecting survivors, supporting communication, and enhancing situational awareness. It is found that current technologies are limited by mobility, autonomy, and communication challenges and insisted on human oversight. The data triangulation insights show that advancing SAR robotics requires technical innovation and their integration into existing emergency-response frameworks. By improved multi-modal human-robot interaction, resilient sensing, partial autonomy, and XR-based training, the rescue efficiency and safety can be improved. It is highlighted that collaboration among technologists, first responders, and policy-makers is essential for validating these systems in real-world conditions. By addressing the highlighted pain points and taking advantage of emerging AI and robotic capabilities, the future SAR mission can be successful in saving lives following catastrophic earthquakes.

CRediT authorship contribution statement

Rajashekhar V S: Writing - original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Gowdham Prabhakar:** Writing - review & editing, Validation, Project administration, Methodology, Investigation, Formal analysis.

Declaration of Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Gemini, ChatGPT and CoPilot in order to understand how they would respond to disasters related questions based on their trained database. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

Relevance to Resilience

Earthquake-induced building collapses pose severe risks to both trapped victims and first responders. This study contributes to resilience by integrating four perspectives namely literature works, public perception, expert opinion, and GenAI-driven insights. This is done to study how robotic systems can strengthen preparedness, response capacity, and decision-making in post-earthquake environments. This work helps in strengthening post-earthquake response capacity, reducing risk to first responders, supporting human-centred and community resilience, and informing resilient technological infrastructure. Therefore, the insights synthesised from literature, public perception, expert experience, and GenAI analysis provide a resilience-focused framework for designing SAR robots that can operate in earthquake environments.

Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships that could have influenced the work reported in this paper.

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The authors have no acknowledgments to declare.

Appendix A. Semi-Structured Interview Questions asked to the Experts

The following are the semi-structured interview questions that were asked of the fire brigades.

Introduction

- Could you briefly introduce yourself and describe your role in the fire brigade?
- How long have you been working in rescue operations?
- Have you participated in or organised any earthquake preparedness training programs for the public?
- What challenges do you face when educating the public about earthquake safety?

Current Situation

- Does your job involve dealing with collapsed building search and rescue operations?
- What are the most common challenges you face during collapsed building search and rescue?
- Have you been trained in collapsed building search and rescue operations?
- Have you done collapsed building search and rescue operations during your service?
- What is the first thing you do when receiving news of a collapsed building?
- How do you reach the spot where the building has collapsed?
- What is the first thing you do after reaching the spot?
- How do you know or find the location of the survivor?
- When do you end the search process?

Current Devices

- What are the current devices that you use to find the survivors?
  - Please list them.
  - Do you need any new devices based on your experience?
- What are the hurdles that you face during the process of rescuing?

Use of Robots:

- Have you ever worked alongside rescue robots during operations?
- Do you think that robots can help or assist you in this process?
- In what process do you expect the robot to assist you?
- What kind of robots do you like to see in searching and rescuing?
- What are the capabilities that you expect the robot to have?
- Have you seen any robots that can help you in the search and rescue process, say on TV or the internet? If so, could you talk about it?

Concluding questions

- What technological advancements, beyond robots, would you like to see in earthquake rescue operations
- How do you think fire brigades and technology developers could collaborate to enhance earthquake rescue operations?
- Thank you for sharing your valuable insights. Is it okay to contact you later if we need further clarification?

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